A Weekly Demand Briefing for Scotland's Emergency Departments

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AI Mastery Capstone: Industry Integrated AI System Synthesis

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28 October 2026

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Overview and Industry Context

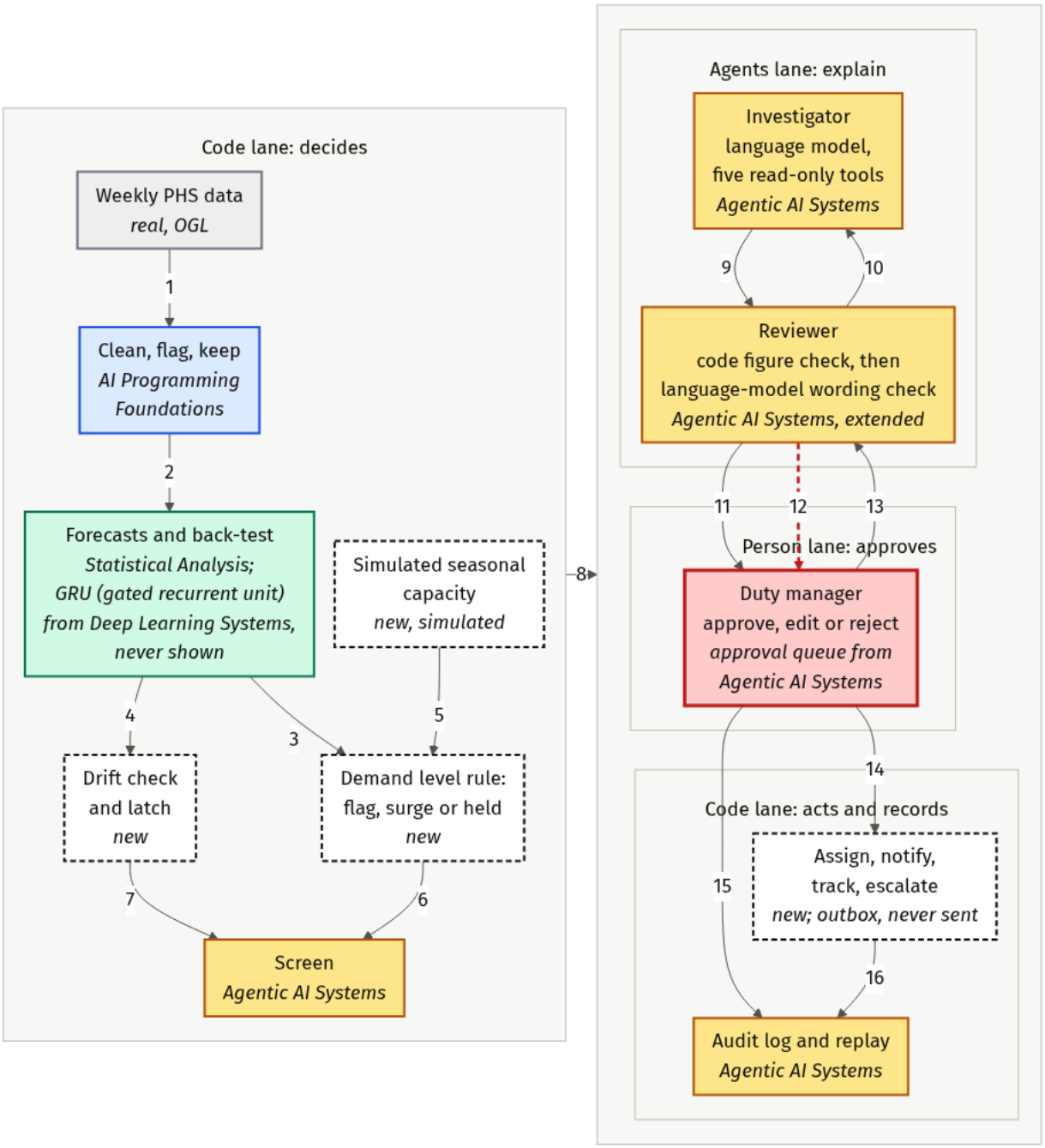
Industry: healthcare operations, unplanned attendances at Scotland's emergency departments (EDs), for the duty managers who staff the front door. Rotas are set weeks ahead: England's National Health Service (NHS) rota code expects resident doctors in training to get rosters at least 6 weeks before a rotation (NHS England, 2025). ED crowding is linked with higher morbidity and mortality (Scottish Government, 2016). A forecast one to four weeks ahead can still inform extra shifts, bank staff or held leave.

At this stage the system gives the duty manager additional information beside their existing process. It replaces no current process and sets no staffing on its own. It needs shadow mode, a run on new weeks with nothing acted on, then a pilot with managers who did not build it and tuning over time, before it could be a primary operational tool.

Why artificial intelligence (AI)? Simple models forecast demand from open data on accident and emergency (A&E) attendances (Public Health Scotland [PHS], 2026), used under the Open Government Licence (OGL) after a SHA-256 fingerprint check. A gated recurrent unit (GRU), a small neural network, runs beside them only as a comparator (Petravić et al., 2025). A duty manager also needs each forecast explained: its range, where the models disagree and how far to trust it. A language model agent drafts that explanation from read-only tools (Yao et al., 2023), while code sets every number (Figure 1).

Figure 1

Code Decides, the Agents Explain, a Person Approves

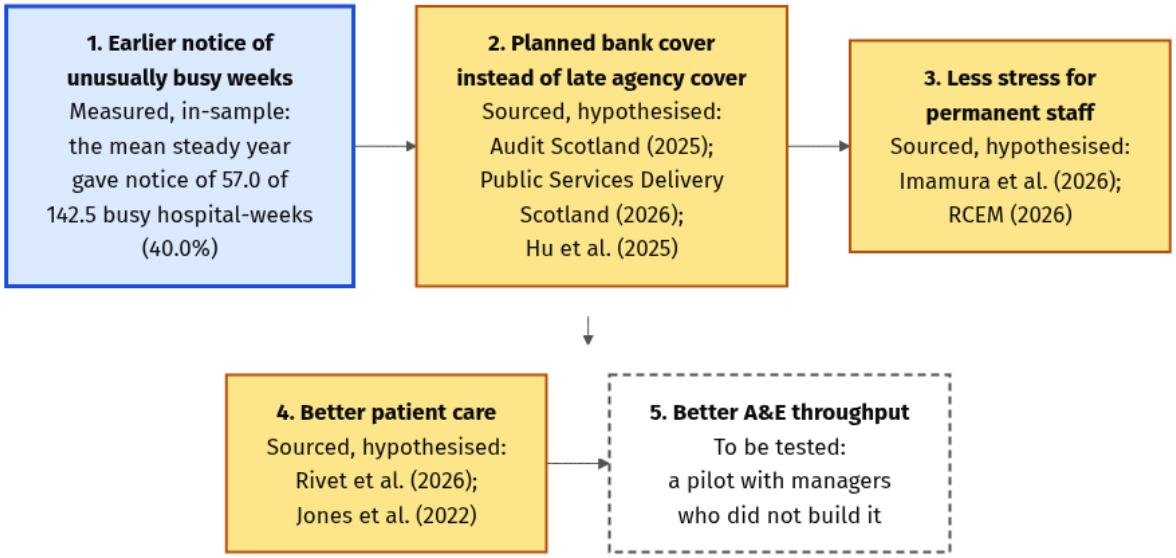


Note. Each box names the prior capstone project it comes from, or marks it new or, for the data, real. Each numbered arrow is a row of the case study's interaction table.

In 2024/25 Scotland spent £256m on agency staff of all kinds and, separately, £353m on the nursing bank (Audit Scotland, 2025, p. 15). Any saving would be mainly the agency premium, with wider value for staff and patients. Figure 2 labels each link of that value chain by its evidence.

Figure 2

The Value Chain, Each Link Labelled by Its Evidence



Note. Link 1 is in-sample, scored on the data the design was tuned on. A hospital-week is busy when its attendances reach 1.05 times simulated capacity, and the mean steady year is the mean of 2019, 2023, 2024 and 2025. Sources: link 2, Audit Scotland (2025), Public Services Delivery Scotland (2026) and Hu et al. (2025); link 3, Imamura et al. (2026) and Royal College of Emergency Medicine (2026), shown in the figure as RCEM; link 4, Rivet et al. (2026) and Jones et al. (2022).

Integration Rationale

I integrated four prior capstone projects (Table 1), spanning data analysis, statistics, deep learning and agentic systems, whose language model brings in generative AI.

AI Programming Foundations set one unexplained merchant's payments aside, flagged, never deleted, and the answer changed without them. So the COVID-19 weeks stay in, flagged, as a real shock to test. Statistical Analysis compared each company with itself in a paired test, so the GRU and the primary model, which drives the flag, are compared on the same weeks. In Deep Learning Systems the best naive forecast put the day's peak in the right 4-hour block more often than either network, so simple models stay the bar to beat. In Agentic AI Systems the Investigator cited figures no tool had returned in 10 cases, and the figure check caught one. So here code writes every figure, and a person approves every brief.

I replay five series, the national total and four hospitals. Outside each series' drift episodes, the weeks its forecasts had stopped fitting, the paired interval favoured the GRU at none of their 20 series-horizon pairs, and the primary at 16. Outside March 2020 to March 2022, it favoured the GRU once. As I stated before the run, the GRU stays a comparator. The frontier backend adds one new technology, Anthropic's client library, behind Agentic AI Systems' model-gateway pattern, chosen by a rule stated before the runs.

Table 1

Prior Capstone Projects, What Each Supplies and Why

Prior capstone project	Method or pattern taken	Component in this system	What it protects, and its cost
AI Programming Foundations	Load, verify, clean, flag and keep	The weekly load, with the COVID-19 weeks kept and flagged	Protects a real shock to test the drift alarm on; costs error scores reported with and without it
Conduct a Statistical Analysis Using Python ("Statistical Analysis")	Intervals, the bootstrap, the trend line and the paired comparison of two models	The rolling-origin back-test and moving-block bootstrap (Hyndman & Athanasopoulos, 2021, Sections 5.10 and 12.5), the flag's error spread and the test of the GRU	Protects against reading a small gap as a real one; costs a resampled interval for every comparison
Deep Learning Systems	Simple forecasts to beat, and a recurrent forecaster of the kind ED arrival research uses (Petravić et al., 2025)	Seasonal naive, the 4-week moving average and the GRU comparator	Protects against trusting a network that a naive forecast matches; costs a yearly refit and three seeds
Agentic AI Systems	The screen, typed read-only tools, the figure check, record and replay, the approval queue and the audit log	The Investigator, the Reviewer, approval and the audit log	Protects every figure a manager reads; costs model calls and a person's time on every brief

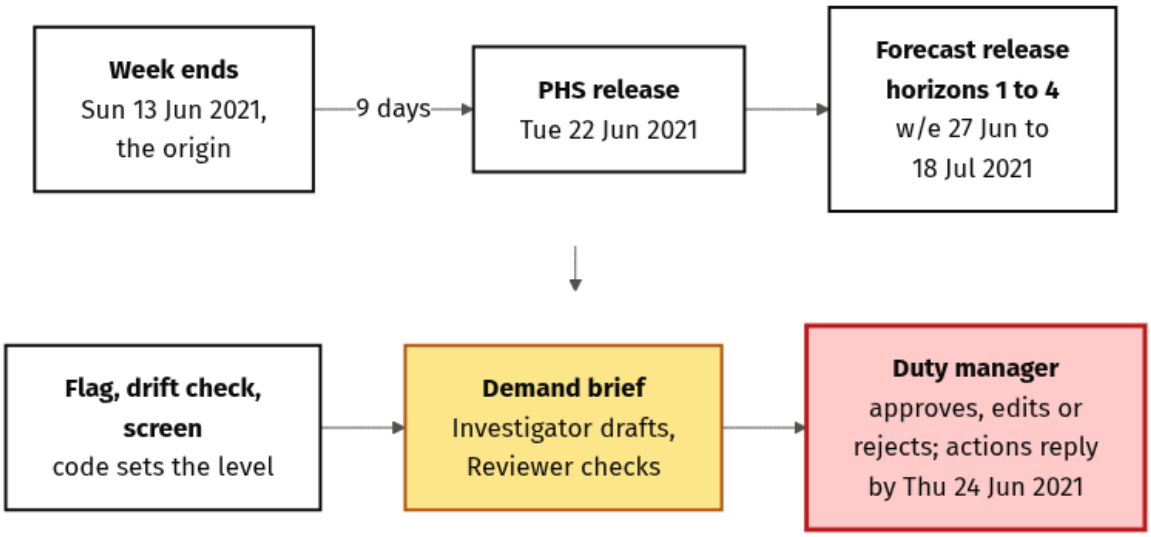
System Design and Technical Decisions

Figure 1 shows three lanes: code decides and then acts, the agents explain, and the duty manager approves, edits or rejects each brief. The data are weekly unplanned A&E attendances at 30 hospitals and nationally, 593 weeks to the week ending (w/e) 11 October 2026. Each week is released 9 days after it ends, so the forecasts cover the four weeks after each release, its release

horizons (Figure 3). After approval, code sets staffing actions, owners and deadlines, and a language model words each notification, kept in an outbox, never sent.

Figure 3

One Week, From the Tuesday Release to the Duty Manager's Decision



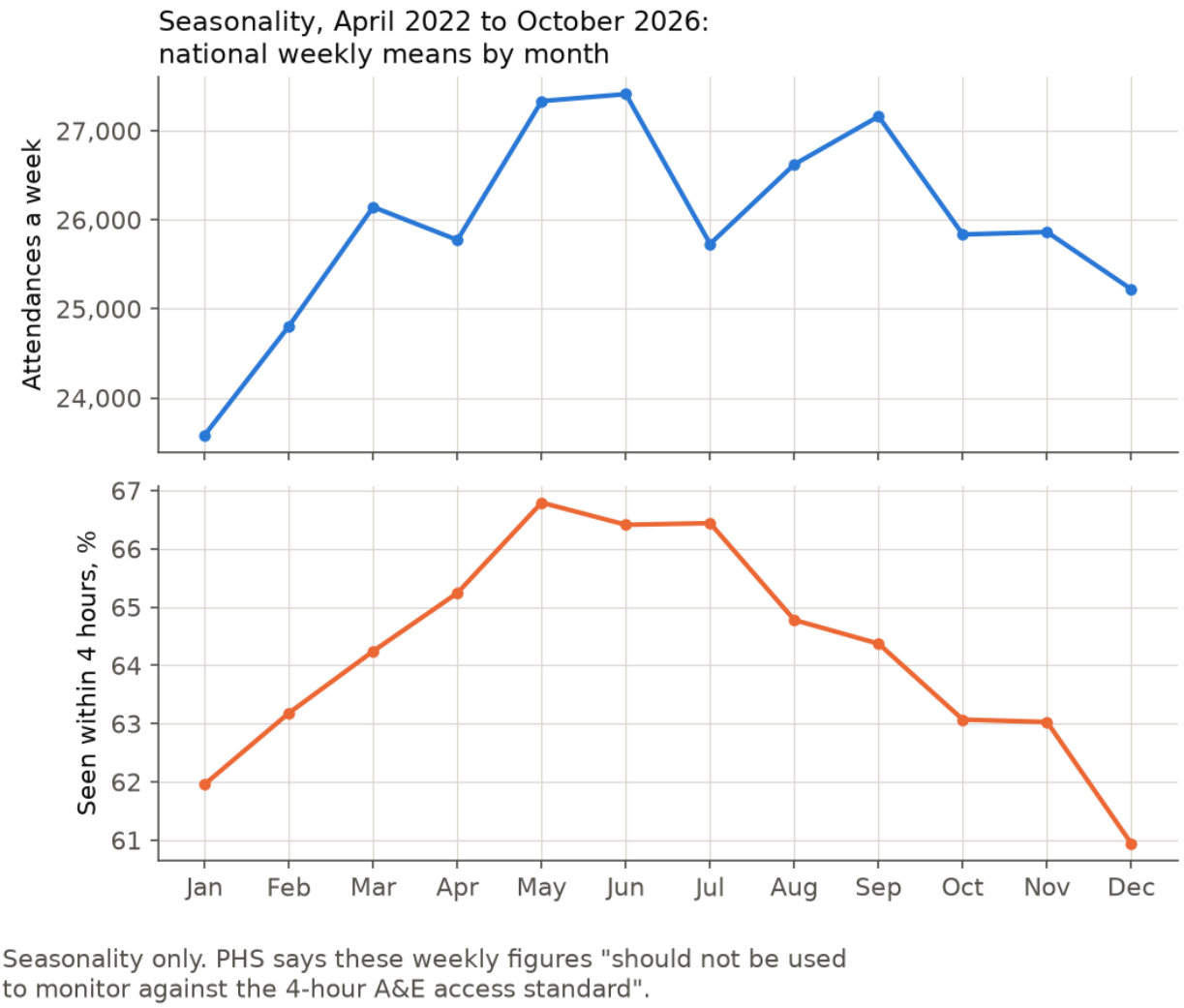
Note. The dates illustrate the release timing for one replay week, Ninewells Hospital's demand week ending 13 June 2021; that week's brief was rejected.

Four simple models forecast each series: seasonal naive, the same week last year, a standard benchmark (Hyndman & Athanasopoulos, 2021, Section 5.2), the 4-week moving average, a linear model and my own seasonally adjusted moving average. The open data hold no beds or staffing, so I use simulated capacity, a seasonal baseline of past demand: the 90th percentile of the hospital's attendances in the same weeks of the previous three years. Code flags an unlatched outlook, one hospital's next four weeks, when the primary model's last 104 errors give any week at least a 50% chance of reaching 1.05 times capacity. The demand level shows demand unusually high for the time of year; it is not pressure, need or priority. It runs from 1 to

4; a flag raises a week to level 2, a raised level. Its bands borrow only the structure of England's Operational Pressures Escalation Levels (OPEL) framework (NHS England, 2024). Attendances peak in May and June, but the 4-hour share is lowest in December and January (Figure 4), so the level measures attendances, not performance.

Figure 4

Monthly Attendances Against 4-Hour Performance



Note. Data from Public Health Scotland (2026). The quotation in the image is from the data set's description (Public Health Scotland, 2026).

The drift alarm fires when attendances fall outside the one-week forecast plus or minus 1.96 standard deviations of its recent errors (Hyndman & Athanasopoulos, 2021, Section 5.5) for two released weeks running. It sets the drift latch, like an alarm that stays on until someone resets it. While it is set, the forecasts are marked untrusted. In a surge, an upward breach, the level comes from the latest released week; after a downward breach the week holds its earlier level. Only a person can reset the latch; in the spot check each reset is simulated. The primary model is then re-selected by its rolling-origin error on the latest 104 released weeks (Hyndman & Athanasopoulos, 2021, Section 5.10). Table 2 sets out the four tradeoffs that shaped the design.

Table 2

Four Tradeoffs That Shaped the Design

Tradeoff	Choice	Gain	Cost
Technical	A simple primary model, with the GRU as a comparator only	A forecast a manager can check in one sentence, with no yearly refit or seeds	Any pattern the simple models miss
Data	Capacity simulated from past demand	A level that can be tested on real attendances	A level that is illustrative, not a statement about beds
Operational	A rare flag	53.8% of flags borne out, against 21.2% of outlooks without one	Only 21.0% of busy outlooks flagged
Model backend	The frontier model, Anthropic's hosted Claude Sonnet 5.5, switchable in `config.json` to a local model run in Ollama	5 of 7 briefs verified, where the local model verified none	A cost per call, an outside service, and a model that cannot be pinned the same way

Note. The flag's figures are in-sample, for the steady years: 2018 from July, 2019 and 2023 to 2026. An outlook or a hospital-week is busy when a week's attendances reach 1.05 times simulated capacity.

The Investigator reasons between tool calls in the ReAct pattern (Yao et al., 2023), writing each figure as a placeholder that code fills. The Reviewer checks every figure against its tool result and screens for the words the Medicines and Healthcare products Regulatory Agency (MHRA) lists as indicative of a medical device (MHRA, 2023, p. 20). A language model then checks the wording. A brief that fails twice is marked "not verified, do not act" and can never be approved.

Ethical Considerations and Responsible AI

Three ethical risks shaped the design.

Automation bias and transparency. Automation bias is a manager approving a brief without weighing it (Goddard et al., 2012). Against it, every brief shows what the level means, its range or why none can be trusted while latched, the models' agreement and the hospital's track record where it has one. The approver, the duty manager, judges each against six criteria written in advance. A person can override every output, following human oversight as a principle, not a legal duty (European Union, 2024, Article 14).

Misuse and status as additional information. The tool is never for ranking hospitals, justifying cuts or monitoring the 4-hour standard. It never sees patient data. Whether it is a medical device rests on my reading of the MHRA's guidance, so it stays an open risk (MHRA, 2023).

Bias by hospital size and accountability. Seasonal naive's median percentage error is 11.0% at sites under 300 attendances a week and 5.4% at 1,300 and over. Raised shares by site ran from 35.7% of outlooks at the Royal Hospital for Children to none at Glasgow Royal Infirmary, both in the largest size band, so the burden varies by site, not only by size. Each brief therefore shows its own site's error, and the level is never read as need. Owners are by role, on

the structure of OPEL's actions by level (NHS England, 2024), and a clinical safety reviewer would sign off before any live use.

Evaluation and Reflection

The eight goals were fixed before the evaluation ran. Every figure is in-sample: the design was tuned on the 2015 to 2026 data that scores it. Error is the mean absolute percentage error (MAPE), as Hyndman and Athanasopoulos (2021, Section 5.8) define it.

What worked. Four goals are met (Figure 5): "forecasts a manager can plan with" (G1), "honest uncertainty" (G2), "shocks are caught" (G3) and "a flag tells a manager something" (G8). Outside the national drift episodes, the primary model's national MAPE is within 1 point of the best simple model at every release horizon, and measurably lower only at the fourth. In the mean steady year the flag alone gave notice of 57.0 of 142.5 busy hospital-weeks (Figure 6). "No unchecked number reaches a manager as approved" (G4) is met, and a figure planted by code was caught. "The agent never sets a level" (G5) holds.

Figure 5

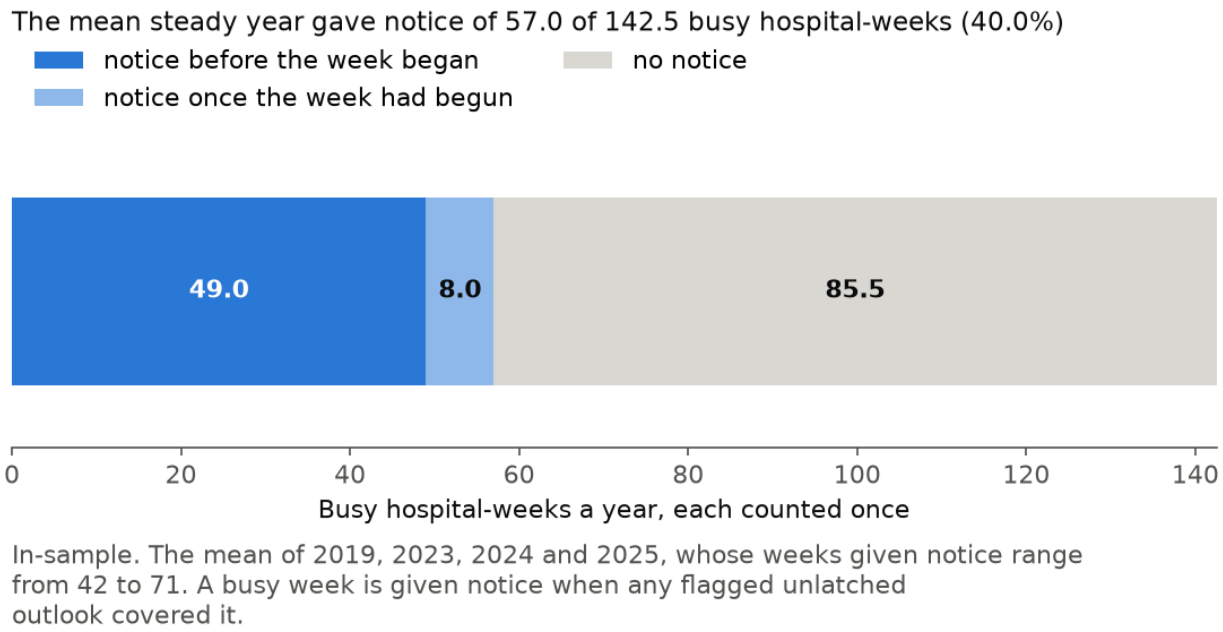
Goals G1 to G8, Each Against Its Threshold

Goal	Met if	Result	Verdict
G1 Forecasts a manager can plan with	within 1 point at every release horizon	-0.18 to -0.74 points against the best simple model	**met**
G2 Honest uncertainty	90% to 98%	95.7% to 96.4% of actuals inside	**met**
G3 Shocks are caught	by Tue 31 Mar 2020; at most 1 a year	known Tue 31 Mar 2020; at most 1 alarm a year	**met**
G4 No unchecked number approved	every approved brief passes; no not-verified brief approved; plant caught	5 of 5 approved pass; 2 not verified, neither approved; plant caught	**met**
G5 The agent never sets a level	always the rule's level	7 of 7 briefs written	**holds**
G6 Briefs are useful to a person	all but one, and Reader B 3 of 4	5 of 7 approved without a factual edit, 6 needed; Reader B 4 of 4 agree	**not met**
G7 Raised levels rare in steady years	under 10% in every steady year	over 10% in 3 years, 10.4% to 11.7%	**narrowly missed**
G8 A flag tells a manager something	at least twice the base rate	2.54 times the base rate, steady years	**met**

G1 to G3, G7 and G8 are in-sample, on 2015 to 2026. G4 to G6 are on the frontier set's seven replay briefs. G5 is an invariant, not a test.

Figure 6

Busy Hospital-Weeks Given Notice in the Mean Steady Year



In-sample. The mean of 2019, 2023, 2024 and 2025, whose weeks given notice range from 42 to 71. A busy week is given notice when any flagged unlatched outlook covered it.

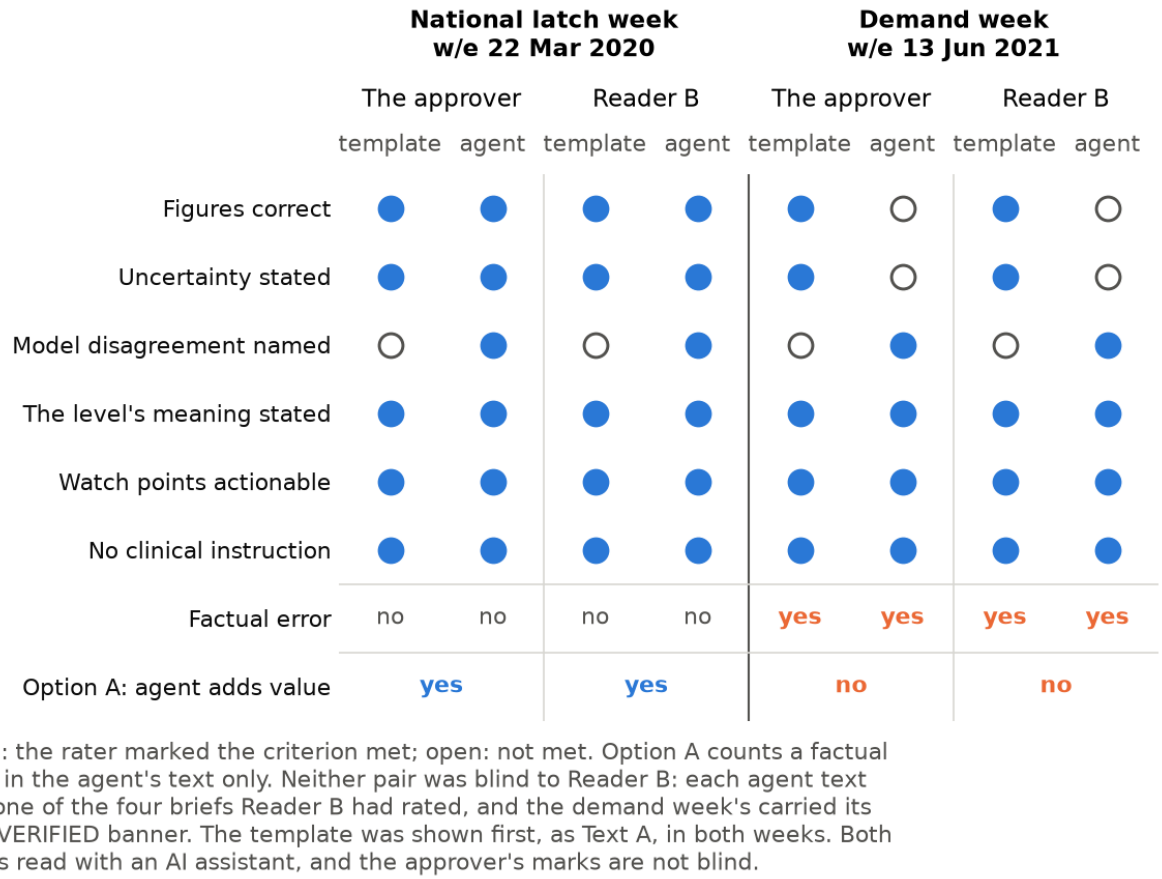
What did not work. The adequacy test, stated before any brief was written, needed 6 of 7 briefs verified and meeting at least 5 of the 6 criteria. The frontier model came closest, at 5 of 7, and the approver chose its set. Because the two refused demand-week briefs could only be rejected, "briefs are useful to a person" (G6) is not met: 5 of 7 were approved without a factual edit, where 6 are needed. Reader B, a second reader using an AI assistant, agreed with the approver's decision on all four briefs, meeting G6's second half, at least 3 of 4. Table 3 qualifies each live set's failures.

The template comparison asked whether the agent's explanation adds value over a fixed template, by a rule fixed before the ratings (Figure 7). In the national latch week it did, for both raters: it addressed the models' agreement, saying every forecast was untrusted while latched, which the template cannot. In the demand week it did not, as that text was the refused brief,

whose trend and fit claims the checks had already stopped. So it adds value overall for neither rater.

Figure 7

Template Against Agent: Both Raters, Both Weeks



Note. Option A is the rule fixed before the ratings: the agent's text adds value in a week when it meets every criterion the template's text meets and at least one more, with no factual error. Both raters marked the demand-week agent text's figures not correct because of its NOT VERIFIED banner, though they had passed the code check.

Table 3

The Three Live Sets of Seven Replay Briefs

Set	Briefs completed	Verified	Cause of each failure	What the safeguard did	Next step
Local model, round 1	3 of 7, stopped at its 840-second budget	0	Every draft broke the text rules on placeholders, units and model names, and a bug in my code resent the first request, so no retry saw what was wrong	The code check refused every draft, so nothing reached approval	Fixed in my code for round 2
Local model, round 2	7 of 7	0	Every draft hit the 8,192-token output cap, which counts the model's thinking	Each brief was marked not verified and could not be approved	A retry with thinking turned off, not yet tried
Frontier model, round 2	7 of 7	5	Both demand-week briefs claimed a rising trend and a good fit that the Reviewer could not trace to a figure in the brief: the history tool did show recent weeks rising, uncited, and the fit claim echoed a line code gave the agent	The wording check refused both, marking them "not verified, do not act", and the approver rejected them	The template now writes demand weeks' explanation, and the shipped template and new runs drop the fit claim; a trend claim that cites one week is a gap to close before shadow mode

Note. The seven replay briefs are real past weeks: Edinburgh's surge (w/e 6 October 2019), the COVID-19 fall at the national series and three hospitals (w/e 22 March 2020), and two demand weeks, at Ninewells (w/e 13 June 2021) and Lorn & Islands (w/e 9 April 2023).

By the comparison's rule, demand-week briefs now take the template's explanation by default, and the agent writes latch, surge and model-disagreement weeks. No threshold for model disagreement was set before the ratings, so that ground keeps no week for the agent; setting one is a next step. The template's no-alarm drift line, which both raters marked a factual error, is versioned. The comparison keeps the rated version 1; the shipped template and new runs say "No drift alarm is recorded; this alone does not establish forecast fit." The recorded runs replay unchanged, and shadow mode tests the default.

"Raised levels are rare in steady years" (G7) is narrowly missed, at 10.4% to 11.7% of outlooks in 2018 (from July), 2023 and 2024, against under 10%. Table 4 qualifies it and two other failures.

Table 4

Other Failure Cases, Each With Its Cause, Safeguard and Next Step

Failure case	Cause	Safeguard	Next step
Raised levels above 10% in 2018, 2023 and 2024 (G7)	My reading, untested: attendances grew past the three earlier years that set capacity	Every raise reached a manager only as a brief to approve, with the hospital's track record	Shadow mode tests the reading before any threshold moves
COVID-19: the linear model's national forecast made at w/e 8 March 2020 missed by 152.8% three weeks later, and every other model missed too	A shock that breaks every model trained on past weeks	The drift alarm confirmed it without warning of it, and the latch held each forecast untrusted until a reset	Shadow mode tests the latch on new weekly releases
No notice of 85.5 of 142.5 busy hospital-weeks in the mean steady year	A rare flag gives up recall for precision	The duty manager's existing process still covers those weeks	Learning what planners use beyond attendance history, within the pilot

An AI independent judge read the seven briefs first and drafted the five edits the approver adopted, with the reset wording corrected, none of them factual. An AI reviewer, not the system's Reviewer, scored the adequacy test. The approver's template marks also used an AI assistant and are not blind. I built the system and played the approver, which the second reader exists to check.

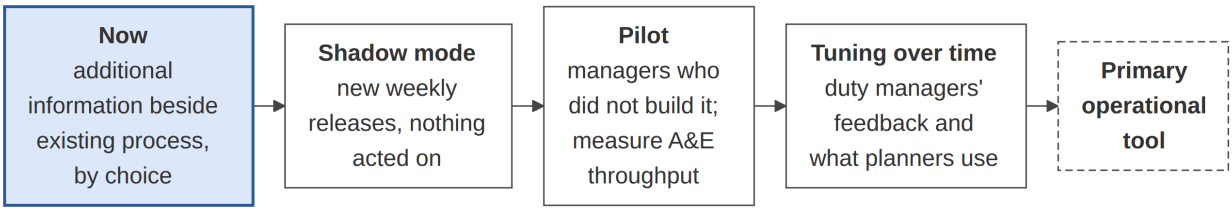
What I learned. Across performance, complexity and responsible use, the GRU's refit and seeds bought no gain, and the frontier model's 5 of 7 verified briefs cost money per call and an outside service. Integration's value lay in the checks between the parts, which stopped both unsupported briefs before approval. The failures showed the agent's limit: it explained the latch and surge weeks well enough to approve, but overreached on both demand weeks, where it had to say why demand was high. I will keep writing each test and its pass mark before the run, and report the result, pass or fail.

Professional Relevance

I kept this a pilot build on purpose: its status as additional information is a choice, not a shortfall. In real use a person still approves every brief before any action follows. Figure 8 shows the path from this pilot build to a primary operational tool.

Figure 8

The Path From Pilot to Primary Operational Tool



This work prepares me for AI solutions architect and machine learning engineer roles. It draws on my enterprise resource planning (ERP) work, where people approve a forecast before

anyone acts on it. The habits it shows, tests stated in advance, a person at every decision and each failure reported with its cause, are what I would bring to an NHS AI team.

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Glossary: every term used in this paper is defined in glossary.pdf, submitted with it.